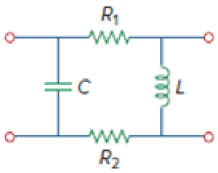
Chapter 19, Problem 19P

Problem

Using Fig. 19.80, design a problem to help other students better understand how to find *y* parameters in the *s*-domain.

Figure 19.80



Step-by-step solution

Step 1 of 10

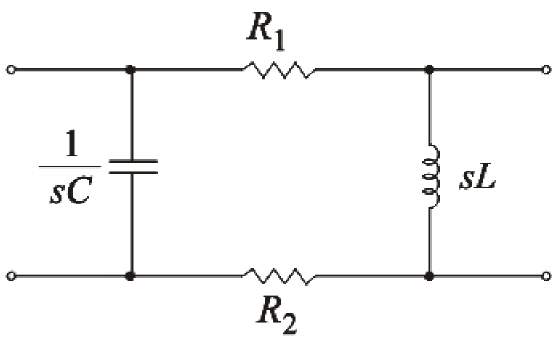
General form of Admittance Parameters

$I_1 = y_{11}V_1 + y_{12}V_2$

$I_2 = y_{21}V_1 + y_{22}V_2$

Consider the given circuit in $s-$domain

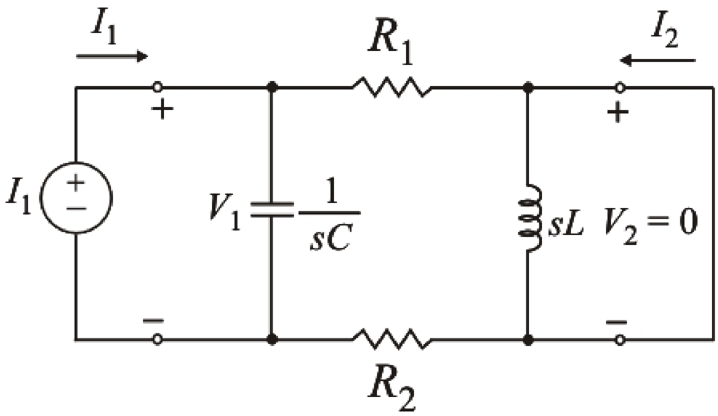
Step 2 of 10



Step 3 of 10

To determine y_{11} and y_{21}, short circuit the output port and connect a current source, I_1 to the input port.

Step 4 of 10



Step 5 of 10

Since sL is short – circuited, then R_1, R_2 are series and parallel to $\frac{1}{sC}$

Hence, $(R_1 + R_2) \| \frac{1}{sC}$

$$\frac{(R_1 + R_2)\frac{1}{sC}}{R_1 + R_2 + \frac{1}{sC}} = \frac{R_1 + R_2}{1 + sC(R_1 + R_2)}$$

$$V_1 = I_1\left(\frac{R_1 + R_2}{1 + sC(R_1 + R_2)}\right)$$

$$\frac{I_1}{V_1} = \frac{1 + sC(R_1 + R_2)}{R_1 + R_2} \quad \cdots\cdots\cdots(1)$$

Step 6 of 10

We know that,

$$y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2 = 0}$$

$$\Rightarrow y_{11} = \left(\frac{1}{R_1 + R_2} + sC\right)$$

Using current division

$$I_2 = -\left(\frac{\frac{1}{sC}}{\frac{1}{sC} + R_1 + R_2}\right)I_1$$

Step 7 of 10

Since from (1),

$$I_2 = \left(\frac{-1}{1+sC(R_1+R_2)}\right)\left(\frac{1+sC(R_1+R_2)}{R_1+R_2}\right)V_1$$

$$\frac{I_2}{V_1} = \frac{-1}{R_1+R_2}$$

We know that,

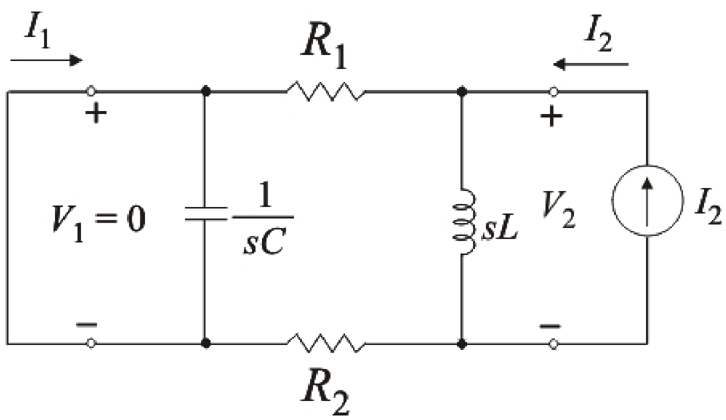
$$y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}$$

$$\therefore y_{21} = \left(\frac{-1}{R_1+R_2}\right)$$

Step 8 of 10

To obtain y_{12} and y_{22}, short – circuit the input port and connect a current source I_2 to the output port. The $\frac{1}{sC}$ is short circuited.

Step 9 of 10



R_1, R_2 are in series and is parallel to sL, then

$$(R_1+R_2)\,||\,sL$$

$$\Rightarrow \frac{(R_1+R_2)sL}{sL+R_1+R_2}$$

$$V_2 = \left[\frac{(R_1+R_2)sL}{sL+R_1+R_2}\right]I_2$$

$$\frac{I_2}{V_2} = \frac{sL+R_1+R_2}{(R_1+R_2)sL} \quad \cdots\cdots(2)$$

Step 10 of 10

We know that,

$$y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$$

$$\therefore y_{22} = \left[\frac{1}{R_1+R_2} + \frac{1}{sL}\right]$$

Using current division, we get

$$I_1 = \left(\frac{-sL}{sL+R_1+R_2}\right) I_2$$

Since from (2),

$$I_1 = \left[\frac{-sL}{sL+R_1+R_2}\right]\left[\frac{sL+R_1+R_2}{(R_1+R_2)sL}\right] V_2$$

$$\frac{I_1}{V_2} = \frac{-1}{R_1+R_2}$$

We know that,

$$y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$$

$$\therefore y_{12} = \left(\frac{-1}{R_1+R_2}\right)$$

Thus, the admittance parameters in matrix form are

$$\boxed{[y] = \begin{bmatrix} \dfrac{1}{R_1+R_2} + sC & \dfrac{-1}{R_1+R_2} \\ \dfrac{-1}{R_1+R_2} & \dfrac{1}{R_1+R_2} + \dfrac{1}{sL} \end{bmatrix}}$$